

**ATTACHMENT 14 – CONTRACTOR TRANSMITTAL (SUBMITTAL NUMBER
723.480-016 A) FROM BECHTEL / PARSONS BRINCKERHOFF (AUTHORIZED
REPRESENTATIVE) TO MODERN CONTINENTAL COMPANY DATED
DECEMBER 27, 1999 INDICATING THE PROPOSED FIX FOR THE FAILED
ANCHOR BOLTS, AS SUBMITTED, IS REJECTED**

(3 pages)



Central Artery/Tunnel

FIELD COPY

CONTRACTOR TRANSMITTAL

TRANSMITTAL DATE:

12/1/99

TRANSMITTAL NO:

CT-337

CONTRACTOR:

Modern Continental

ATTENTION:

Robert Stearns

CONTRACT NO:

9919-COR32

DOCUMENT TYPE

- R. REPRODUCIBLE
- P. PRINT
- C. CATALOG CUT
- E. EQUIPMENT MANUAL
- S. SAMPLE
- T. TECHNICAL DATA OR CALCULATIONS

REVIEW STATUS

- 1. WORK MAY PROCEED.
- 2N. WORK MAY PROCEED SUBJECT TO THE CHANGES INDICATED. RESUBMITTAL NOT REQUIRED.
- 2R. WORK MAY NOT PROCEED ON INDICATED PART(S). RESUBMITTAL REQUIRED. WORK MAY PROCEED ON OTHER PART(S).
- 3. WORK MAY NOT PROCEED. REVISE AND RESUBMIT.
- 4. FOR INFORMATION ONLY.

DATE RECEIVED BY BIPB RE OFFICE:

12-1-99

DATE DUE BACK TO CONTRACTOR:

12-31-99

DATE SENT BACK TO CONTRACTOR:

12-27-99

RESIDENT ENGINEER:

DATE:

12-27-99

ITEM	SUBMITTAL NO. (I.E., 120.010 - 011 - A)		DESCRIPTION	DOC DATE	DOC TYPE	REVIEW STATUS
	SPEC SECT	- SEQ# - REV				
1	723.430	016 A	Adhesive Anchor Repair Procedure		T	2R3
2						
3						
4						
5						
6						
7						

COMMENTS TO CONTRACTOR (USE ADDITIONAL SHEETS AS NECESSARY)

REVIEWER:

DATE:

12/27/99

WHITE - REVIEWER'S COPY GREEN - RESIDENT ENGINEER'S COPY YELLOW - INFORMATION COPY PINK - PROJECT ENGINEER COPY GOLD - CONTRACTOR'S COPY

CRE051 (8/96)

PPM - 1202

December 23, 1999

C09B2 Tunnel Finishes

DR-001 Adhesive Anchors Failures

The DR-001 proposed fix for the failed anchor bolts, as submitted, is rejected.

Prior to beginning the repair process the contractor needs to submit documentation on the manufacturer of epoxy and type being used by Powers Fasteners. i.e., If it is a SIKA product what type is it?

The repair/replacement procedure for the failed anchors is as follows:

1. Remove the failed anchors without damaging the hole or adjacent concrete.
2. Use a carbide bit with hammer drill to re-drill/ream the hole to create a new clean surface. There shall be no foreign matter remaining in the hole.
3. Follow manufacturer's procedures to clean and inspect the hole. Begin grouting from the bottom of the hole working towards the surface or opening of the hole. Install a clean, new anchor and tension test the anchor using 125% of the governing Allowable Tension value from the catalog. For a 5/8" dia. Powerfast adhesive anchor the limiting allowable tension value is the adhesive bond strength. The 125% tension test value is 7900 lbs.
4. Install hanger attachment plate and torque anchor bolts to manufacturer's instructions.

Because of evidence concerning installation problems, i.e. dust in the hole and insufficient adhesive in the hole, the capacity of all of the other anchor bolts is in question and will need to be addressed.

There are two options to resolve this problem.

1. Remove all of the anchors and follow the DR procedures to install new bolts, perform tension test on each bolt, reinstall hanger and torque bolts .
2. Retest all of the anchors using 125% of the governing Allowable Tension value, 7900 lbs .